

PROJECT # 25M066CN

MACC # 25-0442

TITLE: CHILLER& PUMP REPLACEMENT BB270, BB274, BB275 , and HP-121

ATTACHMENTS:

Loop Cleaning Procedures
Spec Sheets for pumps and chiller.

General Description:

Replace chillers at the following buildings.

1. BB-270 75 Ton Chiller and 2- Chilled Water Pumps
 2. BB-274 70 Ton Chiller and 2- Chilled Water Pumps
 3. BB275 75 Ton Chiller and 2- Chilled Water Pumps
 4. HP-121 90 Chiller and 1- Chilled Water Pump
- *Chiller and disconnect sizes should be verified by the contractor.

*** It is the responsibility of the contractor to replace pump/pumps on site of each new chiller installation and assure the pump/pumps will operate the new chiller with the proper GMP. If the existing pump does not meet the new chiller requirements it is the responsibility of the contractor to resize the pump with the proper style and GPM for the chiller. Verifying pump voltages is the responsibility of the contractor.

SCOPE OF WORK:

The contractor shall provide all material, labor, equipment, and supervision required to accomplish the following:

Detailed Requirements and Specifications: The contractor shall make site visits coordinated with PWD, Scotty Sholar-910-389-4418

Chiller Specifications and Basis of Design.

1. All chilled water piping and strainers shall be cleaned prior to installation of new chiller. See attachment for hydronic pipe cleaning and flushing procedures.
2. Replace chilled water pumps and drives on all chiller installs. Pump must meet the manufacturer specifications for each chiller.
3. Chiller can have brazed plate heat exchangers if manufacture can provide a 10-year parts, labor, and refrigerant warranty the entire chiller. In addition to adding a bypass line that allows the chiller to be isolated from the rest of the chilled water piping loop for post-installation/pre-startup system flushing. A factory installed water strainer shall also be included for each chiller.
4. Chiller shall be equipped with a 20amp GFCI convenience outlet for maintenance to be performed. Chiller disconnects shall be replaced along with the flexible liquid tight conduit and connectors between disconnect and the chiller. New disconnect shall meet the manufacturers recommendations. Heat Trace shall protect the evaporator and all piping that is subject to freezing. The 20amp heat trace circuit shall be on a separate circuit. It is the responsibility of the contractor to verify circuit is active. If not, a new circuit shall be run. Location of circuits and existing disconnects listed above.
5. Chiller shall include single point power connection to a non-fused disconnect switch with through-the-door handle and compressor circuit breakers.
6. Chiller condenser fan motors shall include VFD's to optimize chiller efficiency and maintain proper head pressure. The condenser fan motors shall be TEFC PSC style motors.
7. Condenser coils shall have a flexible cationic epoxy polymer e-coat uniformly applied to all metallic surfaces with no material bridging between fins. The electro-coat shall ensure complete HX encapsulation of all conductive surfaces with uniform dry film thickness from 0.6-1.2 mils (15-30um). E-coating shall meet 5B rating for cross-hatch adhesion per ASTM B3359. Corrosion durability will be confirmed through testing no less than 10000 hours salt spray resistance per ASTM B117 using scribed aluminum test coupons. After e-coat cure, HX shall receive a spray applied, 2K polyurethane black topcoat to prevent UV

degradations of epoxy e-coat film. Topcoat shall have a 60-degree gloss (<90%) and dry film thickness of 50-60um.

8. All condenser coils shall be Microchannel coils constructed of aluminum alloy 9150 (Long Life Alloy), to help prevent inter-granular corrosion and increase mechanical strength before and after brazing. Aluminum alloy 3102 coils will not be accepted due to mechanical failure rates. Copper tube and aluminum fin condenser coils will be accepted as an alternate if they can include the factory coating with published 10000-hour salt spray rating listed above.
9. Condenser coils shall be protected by louvered coil guards and compressor housing shall have equivalent protection.
10. Chiller shall sit on Seismic Feet. And rubber pads installed on bottom of rails to elevate side rails to allow wash down of debris from under chiller.
11. Chiller must include thermal dispersion type flow switches. Paddle type will not be accepted. The flow shall be factory installed at the evaporator outlet.
12. Chillers must include phase loss with under/over voltage protection with LED indication of the type of fault to guard against compressor motor burn out.
13. All piping shall be a minimum of schedule 40 mild steel, welded to pipe fitting standards.
14. Emergency chiller connections: a set of emergency tees shall be installed before entering the chiller on supply and return lines. The tees shall match the diameter of the pipe. A butterfly valve and a bolted cover plate shall be installed.
15. Control panel shall have access to microprocessor (controller) for viewing and set up, without interrupting machine operation. Chiller shall include a centrally located weatherproof control panel including DDC microprocessor controller. Control box shall be designed in accordance with NEMA 3R rating. Controllers and screens must be located within the panel and not exposed to sunlight. All wiring within control panel shall be ran through slotted wiring duct with covers. Loose wires with zip ties will not be accepted.
16. Units 75 nominal tons and up shall include reduction mode to allow for reduced sound operation. VFD's are to limit the condenser fan speeds to reduce sound. User shall be able to specify times chiller will operate in sound reduction mode. Chiller shall be set to override the reduced sound mode when the additional capacity is needed. This will make sure the chiller is optimized between required capacity and the lowest sound level possible.
17. Each refrigerant circuit must include a minimum of 2 scroll compressor. Multiple scroll compressors shall be utilized to prevent large capacity loss due to compressor failures. Compressor modules must have isolation valves and controls to lock out individual compressor stages. If low load levels are required, then the chiller is to include hot gas bypass to allow unit operation to 10% of full load to prevent compressors from staging on and off.
18. Each condenser circuit shall have isolation valves, service valves and replaceable core filter driers.
19. Chillers must meet the sound data listed in the below at specified frequencies.

Nominal Tonnage	Octave Band at Center Frequency, 30 ft. from unit.								Overall A-Weighted			
	63	125	250	500	1000	2000	4000	8000	100% Load	75% Load	50% Load	25% Load
30	65	64	61	60	56	51	46	41	61	60	58	57
35	65	64	62	60	56	51	46	41	61	60	58	57
40	65	64	63	61	57	52	47	42	62	61	59	58
45	66	65	64	62	58	52	47	42	63	62	60	59
50	66	66	64	62	58	52	47	42	63	62	60	59
55	66	66	66	62	59	54	49	44	64	63	61	60
60	67	66	67	62	59	54	49	44	64	63	61	60
65	68	67	67	62	60	54	49	44	65	64	62	61
70	68	68	67	62	60	54	49	44	65	64	63	62
75	68	68	68	62	60	54	49	44	65	64	62	61
80	66	63	63	63	62	56	53	53	66	65	63	62
90	67	67	67	64	62	56	54	54	67	66	64	63
100	66	65	65	65	62	56	57	55	67	66	64	63
110	66	67	67	65	61	55	55	55	67	66	64	63
120	68	67	67	65	61	59	60	57	68	67	65	64
130	67	68	68	64	61	60	60	55	68	67	65	64
140	68	69	69	65	61	60	61	56	69	68	66	66
150	69	70	70	65	62	62	62	56	70	69	67	66
160	69	70	70	65	62	63	62	56	70	69	67	66
170	69	71	69	67	64	61	60	58	70	69	67	66
180	69	71	69	67	64	61	61	58	70	69	67	66
190	69	71	69	67	64	61	61	58	70	69	67	66

Sound data must be published to meet or be lower than the above numbers for respective tonnage. Any chiller that do not conform to these published sound targets must provide attenuation to meet at the installing contractor's expense. Sound attenuation should not affect or degrade the chillers performance. The chiller manufacturers catalog data should support all listed sound data.

20. Factory supplied strainer shall be field installed on the evaporator inlet side. There are to be no welds downstream of the strainer prior to the heat exchanger.
21. Base-line flow water must be recorded prior to demo. Post installation water flows must be taken to assure factory required flows are met.

Chiller manufacture must provide "Y" type strainer with the following CV values listed below.

Pipe Size	CV
2"	120
2.5"	165
3:	227
4"	457
5"	641
6"	976
8"	1,607
10"	2,574
12"	3,576

22. Sight glasses shall be installed.
23. Controls shall be BACnet/MSTP, the controller shall be in English, and must monitor phase loss, over and under voltage, line, and load monitoring. Manual latch on critical faults and delayed restart on non-latched errors. Chiller shall enable on 45 Degree F and above.
Any control interface presently installed and operable on the chiller shall be reinstalled and made to operate. This includes hardwire start-stop, alarm, trouble, or integration to a BACnet/MSTP communication bus.
24. Chiller manufacture must have a service organization with at least one service technician within 35 miles from Camp Lejeune, NC.
25. New Chiller must comply with the buy America Act/Free Trade Agreement with the United States of America.
26. Piping diagrams shall be developed by the contractor based on chiller selection.
 - 1.1 All piping shall be approved by PWD/OPS prior to installation.
 - 1.2 All piping shall be minimum of schedule 40 welded steel pipe. All welds shall be primed and painted with a minimum of two (2) coats of alkyd enamel. All external piping shall have heat trace to prevent freezing. All pipes shall be insulated with minimum of 2" of polyisocyanurate insulation. All insulation shall be covered with aluminum UV cover. All piping shall be supported with pipe stands and or welded supports.
27. Electrical feed
 - 1.1 All electrical feeds and controls shall conform to the "NEC". Contractor shall replace electrical disconnects and liquid tight conduit and connectors. Existing wiring can be reused if it meets manufactures recommendations.
 - 1.2 All electrical wiring sizes, voltage and amp draws of equipment is the responsibility of the contractor.
 - 1.3 All information provided in this scope of work is for reference only.
 - 1.4 Site visits shall be required by all bidding contractors.

DETAILED REQUIREMENTS AND SPECIFICATIONS

1. Provide spec sheets for material and equipment being used prior to start of work for government approval. Provide a copy of Maximo Equipment Inventory Sheet for each piece of equipment removed and installed.

SPECIAL SCHEDULING AND ACCESS:

1. Scheduled outages are to be scheduled thru PWD. Outages shall be scheduled fourteen (14) days prior to equipment shut down. Contact Clifton Smith 910-389—8872.

SPECIAL CONDITIONS: Contract shall be completed whit in 300 days after award.

Contractor shall ensure all employees and subcontractors have access and fully vetted.

The contractor shall give Public Works the right to all HVAC equipment, including but not limited to chillers, AHUs, DOAS units, and any DDC controllers that are involved with the sequence of operation for the buildings HVAC system.

The contractor shall give Public Works the right to reclaim all refrigerant from any HVAC system before any equipment is to leave the work area. Any controls that are on existing chiller that need to be installed on new chiller can be removed and installed by the contractor. After Public Works remove any parts as listed above, the chiller will be the property of the contractor to dispose of offsite.

Submittals:

All equipment shall /must be approved. Summit all proposals to the ROICC and PWD for consideration. The contractor shall be required to submit work schedules to POC for approval. The contractor shall submit to the Contracting Officer, the manufacturer's product specifications, complete with technical product data. The product specifications shall, as a minimum, provide technical details on the following items:

- a. Manufacturer's name and product line.
- b. Manufacturer's Warranty

HAZARDOUS MATERIALS: None

ENVIRONMENTAL: None